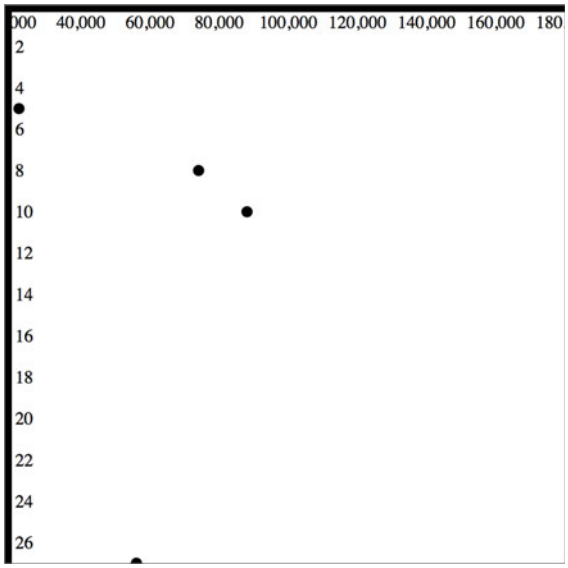


Figure 4.5 The same scatterplot from figure 4.4, but with a pair of labeled axes. The x-axis is drawn in such a way as to obscure one of the points.

we can draw the axis via a selection's `.call()` method from a selection on a `<g>` element where we want these graphical elements to be drawn.

```
var yAxis = d3.svg.axis().scale(yScale).orient("right");
d3.select("svg").append("g").attr("id", "yAxisG").call(yAxis);
var xAxis = d3.svg.axis().scale(xScale).orient("bottom");
d3.select("svg").append("g").attr("id", "xAxisG").call(xAxis);
```

Notice that the `.call()` method of a selection invokes a function with the selection that's active in the method chain, and is the equivalent of writing

```
xAxis(d3.select("svg").append("g").attr("id", "xAxisG"));
```

Figure 4.5 shows a result that's more legible, with the xy positions of the circles denoted by labels in a pair of axes. The labels are derived from the scales that we used to create each axis, and provide the context necessary to interpret this chart.

The axis lines are thick enough to overlap with one of our scatterplot points because the domain of the axis being drawn is a path. Recall from chapter 3 that paths are by default filled in black. We can adjust the display by setting the fill style of those two axis domain paths to `"none"`. Doing so reveals that the ticks for the axes aren't being drawn, because those elements don't have default "stroke" styles applied.

Figure 4.6 demonstrates why we don't see any of our ticks and why we have thick black regions for our axis domains. To improve our axes, we need to style them properly.

4.2.2 Styling axes

These elements are standard SVG elements created by the axis function, and they don't have any more or less formatting than any other elements would when first created.